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# **Identity Security Monitoring Platform: A Detection tool for Privilege Creep**

## **Executive Summary**

Organizations routinely grant employees temporary elevated permissions to complete projects, support business operations, or perform administrative tasks. These permissions are frequently not removed when they are no longer required, resulting in privilege creep, a security issue that increases organizational risk and violates the principle of least privilege. Privilege creep occurs when employees accumulate permissions over time due to role changes, temporary projects, or incomplete deprovisioning, resulting in access that exceeds their current business responsibilities. This commonly occurs when an employee changes roles, transfers to another team, or completes a temporary project.

Privilege creep must be effectively managed because excessive permissions increase an organization's attack surface and can significantly amplify the impact of a compromised account. If a threat actor successfully compromises an over-privileged account, they may gain unauthorized access to multiple systems, sensitive data, and administrative resources beyond what is required for that user's role. Additionally, unmanaged privilege creep may contribute to noncompliance with regulatory and industry standards such as HIPAA, GDPR, or the Sarbanes-Oxley Act. To address these challenges, the proposed solution leverages Microsoft Azure services, including Azure Virtual Machines, Azure Monitor, Log Analytics Workspace, Microsoft Sentinel, Microsoft Sentinel Workbooks, and Kusto Query Language (KQL), to centralize identity monitoring, detect indicators of privilege creep, and support automated security analytics.

Upon completion, the platform will collect Windows Security events, identify privileged account changes, generate alerts for suspicious identity-related activity, and provide dashboards that support security investigations and periodic access reviews once.

From a business perspective, the solution will:

- Improve visibility into privileged account activity
- Detect indicators of privilege creep more efficiently through centralized monitoring
- Reduce manual effort associated with periodic access reviews
- Accelerate investigation of privileged access changes using automated alerts
- Strengthen identity governance and improve audit readiness
- Align with the Zero Trust, MITRE ATT&CK, and NIST CSF 2.0 frameworks

## Company Background

For this project, a fictional organization, generated by artificial intelligence, named Northwind Retail will be used to demonstrate the implementation of an Azure-based Identity Security Monitoring Platform. Northwind Retail is a mid-sized retail and e-commerce company with approximately 500 employees operating across multiple office locations within the United States. The organization supports a hybrid workforce and relies heavily on Microsoft Azure cloud services and Windows Server infrastructure to manage business operations, customer data, and internal information systems.

As the company has expanded, employees have frequently changed roles, participated in cross-functional projects, and received temporary administrative privileges to perform business-critical tasks. While these permissions are necessary to support day-to-day operations, they are not always removed once projects are completed or employees transition into new positions. Over time, users accumulate excessive permissions that exceed the access required for their current responsibilities, creating a condition known as privilege creep.

Privilege creep presents a significant business and security challenge for Northwind Retail. Excessive permission increases the organization's attack surface and elevates the potential impact of a compromised account. If an attacker gains access to an over-privileged account, they may be able to access sensitive business systems, confidential customer information, or administrative resources beyond what the user should be authorized to access. In addition, unmanaged privileges make compliance efforts, security audits, and periodic access reviews more difficult and time-consuming.

To address these challenges, Northwind Retail proposes implementing an Identity **Security** Monitoring Platform built on Microsoft Azure. The platform will centralize Windows security event monitoring, detect indicators of privilege creep using Microsoft Sentinel and Kusto Query Language (KQL), and provide security teams with automated alerts and dashboards to improve visibility into privileged account activity. By adopting this solution, the organization aims to strengthen identity governance, support the principle of least privilege, and reduce the security risks associated with excessive permissions.

## **Business Problem Statement**

Privilege creep occurs when users accumulate permissions that exceed their current job responsibilities. These excessive permissions often result from promotions, department transfer, temporary project assignments, or incomplete deprovisioning processes.

When unnecessary privileges remain assigned, the organizations attack surface expand considerably. A compromised privileged account provides attackers with access to sensitive business systems, confidential information, and administrative functions that would otherwise remain protected. Privilege creep may also complicate compliance efforts and increase audit findings.

To address this challenge, the organization will implement an Azure-based Identity Security Monitoring Platform utilizing Microsoft Sentinel, Azure Monitor, Log Analytics Workspace, and KQL to continuously monitor privileged account activity and detect indicators of privilege creep.

By providing the organization with a platform based around centralized monitoring and automation detection, security teams will be better able to identify excessive permissions, reduce manual reviews efforts, and improve overall identity governance.

## **Project objectives**

Below displays how this platform will commit to business and technical objectives for the company.

### **Business Objectives**

- Reduce organizational risk associated with privilege creep
- Improve visibility into privileged account activity
- Increase compliance and audit readiness
- Strengthening identity governance
- Support least-privilege access management

### **Technical Objectives**

- Deploy an Azure-based security monitoring environment
- Collect Windows Security Events
- Develop KQL detection rules
- Configure Microsoft Sentinel
- Create dashboards for monitoring privileged access
- Generate automated incidents tied to privilege-related events

## Project Scope

### In Scope

- Azure Virtual Machine
- Azure Monitor
- Azure Monitor Agent
- Log Analytics Workspace
- Microsoft Sentinel
- KQL Analytics
- Sentinel Workbooks
- Windows Security Event Monitoring
- Identity-related dashboards

### Out of Scope

- Production Active Directory
- Automated privilege removal
- Microsoft Entra ID PIM implementation
- Multi-cloud monitoring
- Third-party SIEM platforms

## Stakeholder Analysis

Below is an analysis of what each stakeholder in a security setting would likely be interested in and what their duties would be when utilizing the platform.

| Stakeholder                               | Business Interest               | Project Responsibility     |
|-------------------------------------------|---------------------------------|----------------------------|
| Chief Information Security Officer (CISO) | Reduce identity-related risks   | Project sponsor            |
| Security Operations Center (SOC)          | Investigate security incidents  | Monitor Sentinel alerts    |
| IT Administrators                         | Manage privileged accounts      | Implement recommendations  |
| Compliance Officer                        | Meeting regulatory requirements | Review audit evidence      |
| Executive Leadership                      | Reduce organizational risk      | Evaluate business outcomes |

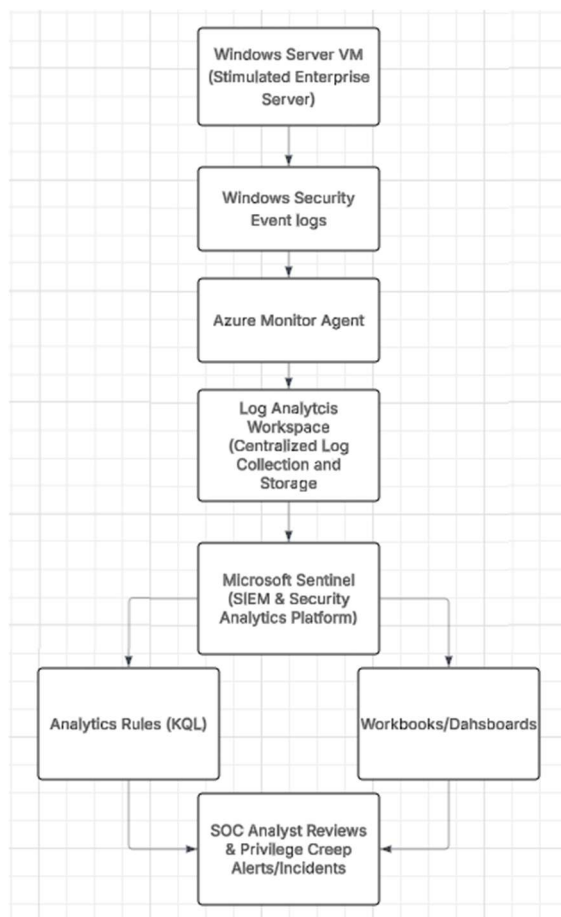
## Business Benefits

The table below displays the business challenges with their proposed solution and expected benefit when the platform is utilized.

| Business Challenge    | Proposed Solution         | Expected Benefit                   |
|-----------------------|---------------------------|------------------------------------|
| Privilege creep       | Continuous monitoring     | Reduced identity risk              |
| Manual log reviews    | Automated Sentinel alerts | Faster investigations              |
| Limited visibility    | Centralized dashboards    | Improved decision making           |
| Excessive permissions | KQL detections            | Better least-privilege enforcement |
| Compliance challenges | Centralized logging       | Improved audit readiness           |

## Architecture Diagram:

Windows Security Events generated by the virtual machine are collected by the Azure Monitor Agent and forwarded to the Log Analytics Workspace. Microsoft Sentinel continuously analyzes these events using Kusto Query Language analytics rules to identify indicators of privilege creep, including privileged group membership changes and elevated logon events. When suspicious activity is detected, Sentinel generated incidents and visual dashboards that enable security analysts to investigate potential identity-related security risks.



**Success Criteria:**

Below are measurable criteria for each objective that is associated with the platform.

| Objective                | Success Metric                                                           |
|--------------------------|--------------------------------------------------------------------------|
| Collect Security events  | Windows Security Events successfully ingested into Log Analytics         |
| Detect privilege changes | Sentinel analytics rules generate alerts for administrator group changes |
| Improve visibility       | Dashboard displays privileged account activity                           |
| Support investigations   | Security incidents created automatically                                 |
| Reduce identity risk     | Platform detects indicators of privilege creep for analyst review        |